KARTIKEYA VEMULA

Analytics Engineer | dbt | BigQuery | GCP Certified | Power BI (PL-300)
vnskartikeya@gmail.com | +1 (443) 630-1157 | LinkedIn | GitHub | Portfolio
Open to relocate across India

PROFILE

Analytics engineer building trusted data systems — dimensional warehouses, tested dbt transformation layers, and decision-ready BI. GCP-certified (Professional Data Engineer, Associate Cloud Engineer) and Microsoft PL-300 certified, with an M.S. in Data Science.

CORE SKILLS

Analytics Engineering: dbt Core, Dimensional Modeling, ELT/ETL Pipelines, Data Quality Testing
Cloud & Warehousing: Google Cloud Platform, BigQuery, Snowflake, PostgreSQL, Docker, GitHub Actions
SQL & Python: SQL, Python, scikit-learn, XGBoost, Anomaly Detection, Root Cause Analysis
BI & Visualization: Power BI, Looker Studio, Streamlit, Plotly Dash
Engineering Practices: Git, CI/CD, A/B Testing, Statistical Analysis, Stakeholder Communication

EXPERIENCE

Graduate Research Assistant, Risk Analytics — UMBC, Baltimore Oct 2025 – Jan 2026
Project: Extreme Value AQI Risk Modeling
- Built an EPA AirData ETL pipeline producing a validated 6,573-record dataset (2006–2023) as the single trusted source for downstream risk modeling.
- Modeled pollutant risk with XGBoost (RMSE 8.99, R^2 0.852), establishing a defensible baseline for expected AQI behavior.
- Quantified a 6.65% conditional tail-risk probability across 98 exceedance events using Extreme Value Theory, delivered as reporting a non-technical monitoring team could act on directly.

Machine Learning Research Assistant — NIT Puducherry, Karaikal May 2023 – Jul 2023
Project: Fish Detection & Recognition System
- Engineered an end-to-end computer vision pipeline in TensorFlow/Keras classifying four fish species as a single reproducible workflow.
- Removed class imbalance by standardizing a 572-image dataset to 143 images per class through augmentation, so accuracy became a meaningful metric.
- Improved generalization on a small dataset using transfer learning with pre-trained CNN architectures, controlling overfitting.

PROJECTS

Instacart Intelligence Platform Python, dbt, PostgreSQL, BigQuery, Docker, Power BI
- Engineered a dimensional warehouse over 3.4M+ orders and 33.8M order-item rows, hardened into 19 dbt models with 238 automated tests passing at zero errors.
- Segmented 206,209 customers into four behavioral cohorts and quantified retention falling to 53.70% by order 10, converting a vague churn concern into a targetable at-risk population.

KPI Reliability & Diagnostic Engine Python, dbt, SQL, Streamlit, GitHub Actions
- Benchmarked three anomaly-detection approaches against labeled ground truth and cut false positives from 2,958 to 563 (−81%) while recall held at 100%.
- Built contribution-share root cause analysis that isolated one region-device-browser segment as 100% of a 44.6% conversion-rate shortfall, plus a 9-check data quality gate blocking inference on failure.

Customer Intelligence Data Warehouse Python, dbt, PostgreSQL, Docker, Power BI
- Converted noisy retail transaction data into 14 tested dbt models (41 passing tests) governing £3,279,364.61 in revenue across 5,047 customers.
- Exposed UK revenue concentration at 84.86% of total and a 67.25% repeat-purchase rate across 3,394 buyers, shaping retention and geographic-dependency priorities.

CERTIFICATIONS

Google Cloud Certified: Professional Data Engineer • Google Cloud Certified: Associate Cloud Engineer • Microsoft PL-300: Power BI Data Analyst Associate • dbt Fundamentals • Snowflake Data Warehousing Workshop • Cisco Data Analytics Essentials • Kaggle Advanced SQL • Google Analytics Certification

EDUCATION

M.S. Data Science — University of Maryland, Baltimore County (UMBC), USA 2024 – 2026
GPA 3.78 / 4.0 | Focus: Machine Learning, Statistical Modeling, Experimental Design, Time Series Analysis

B.Tech Computer Science (AI & ML) — GITAM University, India 2020 – 2024
8.89 GPA, First Class with Distinction

EXTRA-CURRICULAR & LEADERSHIP

- Mentored peers in Mathematics, Digital Logic, Data Communication, and Data Analysis & Machine Learning.
- Led a 3-part UMBC peer-teaching series on ML terminology, model training, and real-world case studies.